

1.Create a Data Frame with three columns namely name,age,city.

Program:

```
import pandas as pd

data = {
    "Name": ["x", "y", "z", "a", "b"],
    "Age": [23, 20, 18, 19, 24],
    "City": ["Chennai", "Madurai", "Trichy", "Vellore", "Kanyakumari"]
}

df = pd.DataFrame(data)

print(df)

print("\n")
```

Output:

	Name	Age	City
0	x	23	Chennai
1	y	20	Madurai
2	z	18	Trichy
3	a	19	Vellore
4	b	24	Kanyakumari

2.Filter the row where Age>20

Program:

```
r = df[df["Age"] > 20]

print(r)

print("\n")
```

Output:

	Name	Age	City
0	x	23	Chennai
4	b	24	Kanyakumari

3.Add a column named fees

Program:

```
df["Fees"] = [10000, 20000, 30000, 40000, 50000]
print(df)
print("\n")
```

Output:

	Name	Age	City	Fees
0	x	23	Chennai	10000
1	y	20	Madurai	20000
2	z	18	Trichy	30000
3	a	19	Vellore	40000
4	b	24	Kanyakumari	50000

4.Update column values increases the fees by 20%

Program:

```
df["Fees"] = df["Fees"] * 1.20
print(df)
print("\n")
```

Output:

	Name	Age	City	Fees
0	x	23	Chennai	12000.0
1	y	20	Madurai	24000.0
2	z	18	Trichy	36000.0
3	a	19	Vellore	48000.0
4	b	24	Kanyakumari	60000.0

5.Drop a colum city

Program:

```
df = df.drop(columns=["City"])  
  
print(df)  
  
print("\n")
```

Output:

	Name	Age	Fees
0	x	23	12000.0
1	y	20	24000.0
2	z	18	36000.0
3	a	19	48000.0
4	b	24	60000.0

6.Rename column Age=>Years

Program:

```
df = df.rename(columns={"Age": "Years"})  
  
print(df)  
  
print("\n")
```

Output:

	Name	Years	Fees
0	x	23	12000.0
1	y	20	24000.0
2	z	18	36000.0
3	a	19	48000.0
4	b	24	60000.0

7.Sorting by fees

Program:

```
df = df.sort_values("Fees", ascending=False)

print(df)

print("\n")
```

Output:

	Name	Years	Fees
0	x	23	12000.0
1	y	20	24000.0
2	z	18	36000.0
3	a	19	48000.0
4	b	24	60000.0

8.Grouping and Aggregation(Group by Dept and Avg of Fees)

Program:

```
df["Department"] = ["CSE", "IT", "CSE", "ECE", "IT"]

avg_fees = df.groupby("Department")["Fees"].mean()

print(avg_fees)

print("\n")
```

Output:

Department	
CSE	48000.0
ECE	24000.0
IT	30000.0

Name: Fees, dtype: float64

9.Merge two data frames

Program:

```
import pandas as pd

df1 = pd.DataFrame({
    'ID': [1, 2, 3, 4],
    'NAME': ['MANI', 'PRIYA', 'RAVI', 'KUMAR']
})

df2 = pd.DataFrame({
    'ID': [1, 2, 3, 4],
    'Sum': [450, 380, 500, 420]
})

df = pd.merge(df1, df2, on='ID')

print(df)
```

Output:

	ID	NAME	Sum
0	1	MANI	450
1	2	PRIYA	380
2	3	RAVI	500
3	4	KUMAR	420

10.Handling Missing Values

Program:

```
import pandas as pd

import numpy as np

data = {

    'Name': ['A', 'B', 'C', 'D'],

    'Age': [20, np.nan, 22, 21],

    'City': ['Chennai', 'Madurai', np.nan, 'Trichy']

}

df = pd.DataFrame(data)

print(df)

print(df.isnull().sum())

df['Age'] = df['Age'].fillna(df['Age'].mean())

print(df)
```

Output:

```
Name Age  City
0  A  20.0  Chennai
1  B   NaN  Madurai
2  C  22.0    NaN
3  D  21.0  Trichy

Name  0
Age    1
City    1

dtype: int64

Name Age  City
0  A  20.0  Chennai
1  B  21.0  Madurai
2  C  22.0    NaN
```

